Two grind options on AMS-AGY v3

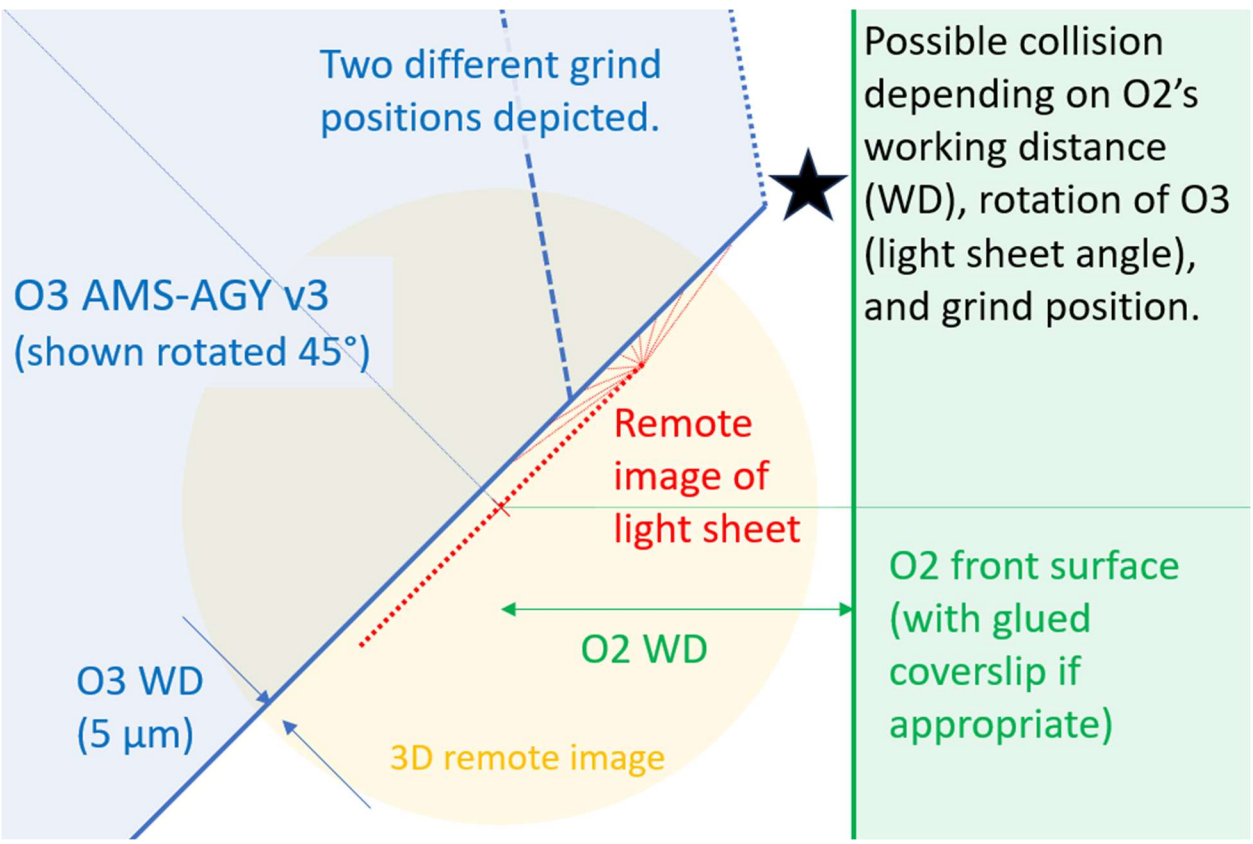
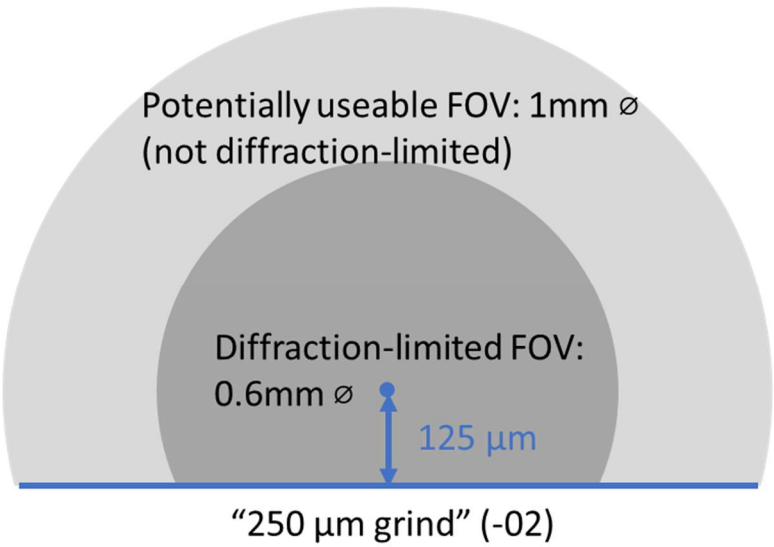
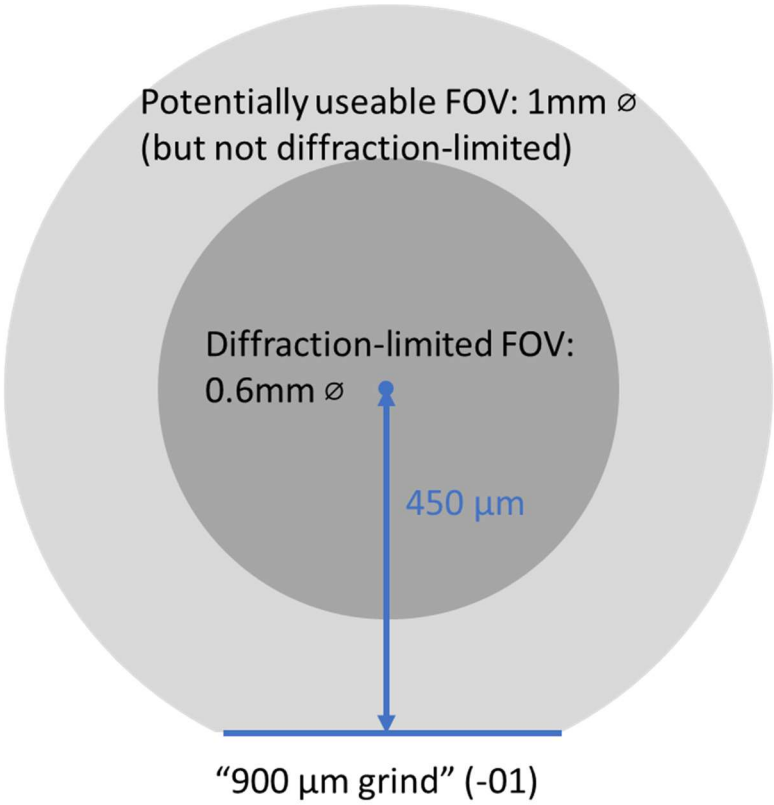
There are two variants of the 57-14-7 AMS-AGY v3 objective. The only difference is the placement of the grind which allows the AMS-AGY objective to be co-focused with the O2 objective. We call them "250 µm grind" and "900 µm grind" variants, designated on paperwork with internal PNs ending in -02 or -01 respectively. **As a general rule, customers using a 40x/0.95 objective as O2 will prefer the "250 µm grind" variant, and all other customers will prefer the "900 µm grind" variant.**

Due to the short working distance of 40x/0.95 objectives, in order to co-focus with the AMS-AGY v3 objective part of its optical FOV needs to be sacrificed. In the "250 µm grind" variant, the optical FOV is truncated along a line which passes 125 µm from the center of the objective. Along the other directions the full FOV remains.

In the "900 µm grind" variant, the glass removed with the grind falls outside of the diffraction-limited FOV. This version can co-focus 20x/0.8 O2 objectives from Nikon, Olympus, and Leica. In this variant the glass is truncated such that the semi-FOV is 450 µm instead of 125 µm.

Due to the non-zero WD, the grind positions are 35 µm larger mechanically than optically (160 µm and 485 µm from the mechanical centerline).

This diagram depicts the possible collision of O2 and O3 when O3 is rotated about its centerline, pushing the truncated portion of the AMS-AGY objective toward O2:



The next diagrams show a view of the objective's FOV from on end, indicating where the glass is truncated for the two variants. Note that the "250um grind" variant truncates the diffraction-limited FOV.